

computer vision task 2.

2.12 The targets are presented in a CSV format.

- The file links a frame (.Tpg file) to a class id.

Difference between training and validation dataset are:

- Training dataset are fed to the model, it is what the model learns and from then adjust its weights/biases to fit and increase the accuracy when it is tested with validation dataset.
- Validation dataset is the dataset used to test the trained model. This dataset has its own correct value so when the model guesses wrong, we can know and calculate its accuracy and from then improve the model.

2.27 There are 7 convolutional layers and only the first 6 have pooling layers.

⇒ There are 6 convolutional "blocks"

The layers they are built from are:

- Conv2D layer: Extract features.
- Batch Normalization layer.
- Function layer: To calculate output or reduce it to some specific range like from -1 to 1.
- Pooling layer reduces spatial dimension ⇒ ~~code~~ model runs efficiently.

2.37 The 2 attributes of the loss function observer:

- One is for class prediction which is classification.
- Another is for box coordinate prediction (localization)

Refine:

- classification: This helps us know how accurate are the model's prediction based on what category is an object classified into.
- localization: This measures the difference between predicted boundary box and ground truth box.